Ratios and Equivalent Ratio Tables
Answer Key

Grade 6

Quick Answers

1. word form = $\frac{3}{5}$, colon form = $\frac{3}{5}$, fraction form = $\frac{3}{5}$
2. colon form = $\frac{4}{7}$, fraction form = $\frac{4}{7}$
3. (a) = $\frac{5}{8}$, (b) = $\frac{3}{8}$
4. =
5. $\frac{4}{5}$
6. 12
7. 25
8. (a) = $\frac{3}{5}$, (b) = $\frac{5}{8}$
9. $\frac{5}{8}$, $\frac{7}{10}$, $\frac{3}{4}$
10. number who walk = 10, —

What is verified

9 of 10 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problem 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 6
6.RP.A.2 – *problems 1, 2, 3, 5, 8*
6.RP.A.3 – *problems 4, 6, 7, 9, 10*
Difficulty 1–4 of 5 across 16 tagged checks.

Common wrong answers

6. *If they got 3: read the ratio numbers as the actual counts, answering 3 red marbles*
6. *If they got 30: used the part-to-part fraction 3/2 times the total instead of 3/5 of the total*

1. Write the ratio of shaded to unshaded counters three ways (3 shaded, 5 unshaded in the picture).

Solution: Count the picture: 3 shaded counters and 5 unshaded counters. The ratio of shaded to unshaded puts the shaded count first. All three forms name the same comparison:

$$\boxed{3 \text{ to } 5 \;=\; 3{:}5 \;=\; \frac{3}{5}}$$

2. A fruit bowl holds 4 apples and 7 oranges. Write the ratio of apples to oranges in colon form and in fraction form.

Solution: Apples are named first, so 4 goes first in the ratio: 4 apples to 7 oranges. In both forms the order must stay apples, then oranges. $\boxed{4{:}7 \;=\; \frac{4}{7}}$

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3. Shaded and unshaded tiles (5 shaded, 3 unshaded): (a) shaded to all tiles, (b) unshaded to all tiles.

Solution: These are part-to-**whole** ratios, so first find the whole: $5 + 3 = 8$ tiles in all.

(a) shaded to all tiles: 5 out of 8. $\boxed{\frac{5}{8}}$

(b) unshaded to all tiles: 3 out of 8. $\boxed{\frac{3}{8}}$

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4. Are 6:8 and 9:12 equivalent ratios? Write $=$ or $\neq$.

Solution: Simplify each ratio by dividing both parts by their greatest common factor:

$$\frac{6}{8} = \frac{6 \div 2}{8 \div 2} = \frac{3}{4} \qquad\qquad \frac{9}{12} = \frac{9 \div 3}{12 \div 3} = \frac{3}{4}$$

Both simplify to the same ratio, so they are equivalent. $\boxed{\frac{6}{8} = \frac{9}{12}}$

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5. Mr. Lee's class has 12 girls and 15 boys. Write the ratio of girls to boys as a fraction in simplest form.

Solution: Girls first: $\frac{12}{15}$. The greatest common factor of 12 and 15 is 3, so divide both parts by 3:

$$\frac{12}{15} = \frac{12 \div 3}{15 \div 3}$$

which gives the simplest form. $\boxed{\frac{4}{5}}$

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6. Red and blue marbles in the ratio 3:2, 20 marbles in all. How many are red?

Solution: The ratio 3:2 makes $3 + 2 = 5$ equal parts. Divide the total into 5 parts: $20 \div 5 = 4$ marbles per part. Red gets 3 parts: $3 \times 4 = 12$. $\boxed{\text{12 red marbles}}$

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7. Fill in the missing value: $\dfrac{5}{8} = \dfrac{?}{40}$.

Solution: Ask what the denominator was multiplied by: $8 \times 5 = 40$. Equivalent ratios multiply **both** parts by the same number, so the numerator is $5 \times 5 = 25$. Check: $\frac{25}{40} = \frac{5}{8}$ after dividing both parts by 5. $\boxed{25}$

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8. A shelter cares for 6 dogs and 10 cats: (a) dogs to cats, (b) cats to all animals, each in simplest form.

Solution: (a) Dogs to cats is part-to-part: $\frac{6}{10}$, and dividing both parts by 2 gives the simplest form. $\boxed{\frac{3}{5}}$

(b) Cats to all animals is part-to-whole. The whole is $6 + 10 = 16$ animals, so the ratio is $\frac{10}{16}$; divide both parts by 2. $\boxed{\frac{5}{8}}$

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9. Order 5:8, 3:4, 7:10 from least to greatest.

Solution: Rewrite each ratio as a decimal to compare:

$$\frac{5}{8} = 0.625 \qquad \frac{3}{4} = 0.75 \qquad \frac{7}{10} = 0.7$$

Order the decimals from least to greatest: $0.625 < 0.7 < 0.75$, and translate back to the ratios. $\boxed{\frac{5}{8},\ \frac{7}{10},\ \frac{3}{4}}$

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10. 25 students, walk to bus ratio 2:3. Find how many walk, and explain why the 2 does not mean exactly 2 students.

Solution: The ratio 2:3 makes $2 + 3 = 5$ equal parts, and $25 \div 5 = 5$ students per part. Walkers get 2 parts: $2 \times 5 = 10$. $\boxed{\text{10 students walk}}$

Explanation (instructor judges): A full-credit explanation says that a ratio compares the *relative* sizes of the groups, not the exact counts: out of every 5 students, 2 walk and 3 ride. Since the class holds 5 groups of 5 students, both parts of the ratio scale by 5, giving $2 \times 5 = 10$ walkers, not 2. Sam read the ratio as a count instead of a comparison.